

Ideation Phase

Define the Problem Statement

Date	10/07/2026
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Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Problem Statement (Customer Problem Statement Template):

Educational institutions and training organizations often evaluate students based on written examinations. However, assessing whether a student truly understands a concept through spoken explanations is difficult, time-consuming, and subjective.

Teachers must manually listen to every student's explanation, identify whether important concepts were covered, evaluate communication skills, and provide feedback. This manual process becomes inefficient when the number of students is large and may lead to inconsistent evaluations.

There is therefore a need for an intelligent AI-based system that can automatically analyze spoken explanations, evaluate conceptual understanding, measure speech fluency, and generate structured feedback.

The proposed project develops a Voice-Based Concept Understanding Analyser (VBCUA) using Artificial Intelligence, Speech Recognition, Natural Language Processing, and Audio Signal Processing. The system converts speech into text using OpenAI Whisper, compares the explanation with a reference concept using Sentence-BERT semantic similarity, analyzes fluency using Librosa, and generates AI-powered feedback using Google's Gemini model. The application also creates downloadable PDF reports for performance tracking.

Example for: Voice-Based Concept Understanding Analyser (VBCUA)

I am	I'm trying to	But	Because	Which makes me feel
A student	Explain a concept through speech	Teachers cannot accurately evaluate every explanation manually	Teachers cannot accurately evaluate every explanation manually	Unsure whether my conceptual understanding and communication skills are good enough

